

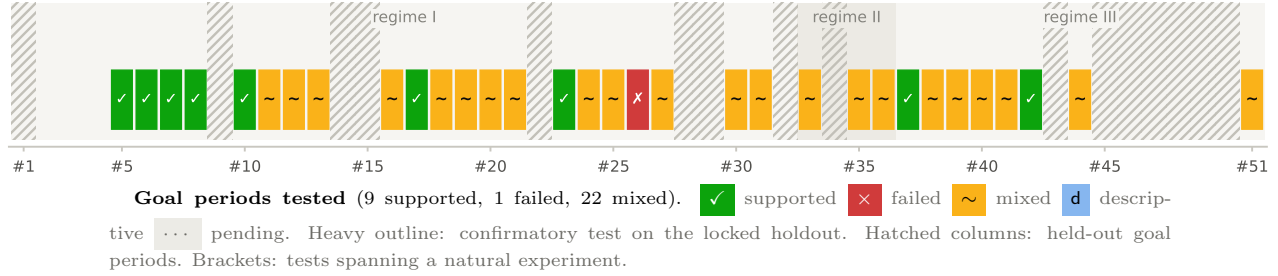
H34: Idea cascades follow a power law whose exponent reads off the distance to criticality

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04 (round 1b)

Complete 50% Faithfulness 2/5 Usefulness 2.5/5

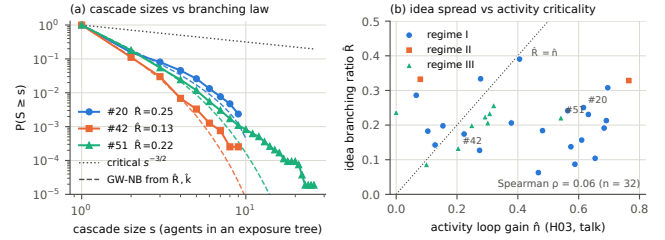
Bottom line. Idea cascades in the village are strongly subcritical ($\hat{R} \approx 0.2$), exposure-driven and heterogeneous rather than power-law. Branching ratio, not exponent or activity loop gain, is the number that forecasts how far an idea spreads.

level: hypothesis · pipeline checks 86% · updated 2026-10-04 (round 1b)



WHAT HAPPENED

We hashed novel terms, URLs, numbers and names in 32 non-holdout goal periods (132k ideas). Each agent's first use was parented to the latest visible earlier use, giving exposure trees. Ideas spread subcritically everywhere: $\hat{R} = 0.06\text{--}0.39$, all CIs below 1. A new idea reaches a second agent 6–24% of the time. Round 1b (context-ledger visibility): adoption is time-locked to recent visible uses beyond a common field in 32/32 periods (round 1: 27/32; regime I was mismeasured). But uses not yet read predict adoption at about half that strength: part of the lock is convergence, not copying. Pure $s^{-3/2}$ is rejected in all 32; tails are heavier than one branching law. Leaders are not idea sources. Failed: activity loop gains do not predict spread; the pre-registered forecast was overconfident. Holdout not run.



Observables. (a) Fraction of exposure trees reaching at least s agents in #20, #42 and #51, with the branching law predicted from $\hat{R}, \hat{k}$ (dashed) and the critical $s^{-3/2}$ (dotted). Cascades are far from critical, and somewhat heavier-tailed than one branching law. (b) One dot per period. Idea branching $\hat{R}$ is unrelated to the activity loop gain $\hat{n}$ from H03; most periods lie below $\hat{R} = \hat{n}$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The usable output is a two-number forecast. Measure $\hat{R}$ (first uses with a visible parent, over all first uses) on the last two days of logs. At the village's $\hat{R} \approx 0.2$, expect 83% of new ideas or claims to stay with their author and about 1 in 150 to reach five agents. With day drift added, this post-hoc rule reaches 70–84% coverage at 90% nominal; it is frozen for the holdout. The negative lessons matter more. Activity criticality (Hawkes n) says nothing about idea spread. In a shared room, “who saw it first” cannot separate copying from independent invention: compare read with not-yet-read uses at the same lag. A power-law exponent is just $\hat{R}$ in disguise. Nothing is near runaway: misinformation here dies locally unless it is an artifact everyone must touch.

PHYSICAL MODEL

Contagion (model 03) as a Galton–Watson cascade (09) in a room of N agents. Adopters have negative-binomial offspring with mean R (branching ratio) and dispersion k . s : tree size. s_c : cutoff. h_k : adoption hazard per talk turn with k recent visible sources. R_c : contagion share of $\hat{R}$. η : daily drift of R in the forecast rule.

$$\hat{R} = \frac{\# \text{ first uses with an agent parent}}{\# \text{ agent first uses}}, \quad R_c = \hat{R} (1 - h_0/h_{\geq 1})$$

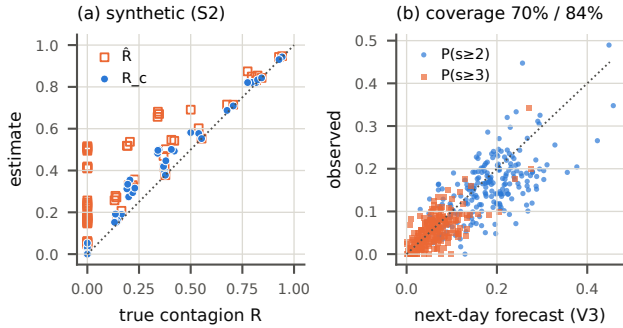
$$P(s | R, k) = \frac{\Gamma(ks + s - 1)}{\Gamma(ks)\Gamma(s + 1)} \frac{(R/k)^{s-1}}{(1 + R/k)^{ks+s-1}} \sim s^{-3/2} e^{-s/s_c}$$

$$s_c^{-1} = R - 1 - \ln R, \quad R_d = \hat{R} e^\eta, \quad \eta \sim \mathcal{N}(0, 0.25^2)$$

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 subcritical, $\hat{R} < 1$	$\hat{R}$ 0.06–0.39, CI < 1 in 32/32 periods, 113/113 class cells	pass
P1 order $U > N > D$	artifacts 0.35 > numbers 0.27 > names 0.19	fail
P2 $\hat{R} < \text{activity } \hat{n}$ (HH108)	25/32 periods	pass
P3a branching-law tail	$P(s \geq 3)$ in band 27/32; $P(s \geq 5)$ heavier in 25/32	fail
P3b no $s^{-3/2}$; $\tau_{\text{app}} \geq 2$	rejected 32/32; τ_{app} 2.4–4.3, $\rho(\tau_{\text{app}}, \hat{R}) = -0.95$	pass
P4 $\hat{R}$ tracks $\hat{n}$	$\rho = 0.06$ (Holm $p = 1$)	fail
P5 contagion beyond field	HR ₁₀ CI > 1: 27/32 → 32/32 (1b); regime I 2.3 → 6.4	pass
1b: read vs unread uses (H57)	read > unread 30/32, but unread CI > 1 in 25/32 (ratio 0.56)	half convergence
1b native: leaders seed spread	#35 ratio 1.11 [0.82, 1.45]; #26 0.36 [0.27, 0.46]	fail
P6 simple contagion	HR(2 vs 1) bracket 0.5–4.2	ambiguous
P7 forecast coverage $\geq 80\%$	36%/46%; post hoc V3 70%/84%	fail



Second figure. (a) On real village timelines with synthetic ideas, a common field inflates $\hat{R}$ (squares) even with no contagion; R_c (dots) recovers the true contagion R . (b) The post-hoc rule's next-day forecasts of $P(s \geq 2)$, $P(s \geq 3)$ vs observed, over 219 period-days.

GOAL PERIODS IN DETAIL

G05 ✓ 1b: supported (ledger visibility: R 0.22, HR₁₀ 5.8 [...]
G06 ✓ 1b: supported (ledger visibility: R 0.11, HR₁₀ 5.6 [...]
G07 ✓ 1b: supported (ledger visibility: R 0.10, HR₁₀ 16.1 ...
G08 ✓ 1b: supported (ledger visibility: R 0.13, HR₁₀ 1.7 [...]
G10 ✓ 1b: supported (ledger visibility: R 0.19, HR₁₀ 9.7 [...]
G11 ~ 1b: mixed (ledger visibility: R 0.21, HR₁₀ 12.2 [6.7...
G12 ~ 1b: mixed (ledger visibility: R 0.28, HR₁₀ 6.2 [4.7,...
G13 ~ 1b: mixed (ledger visibility: R 0.15, HR₁₀ 4.4 [3.0,...
G16 ~ 1b: mixed (ledger visibility: R 0.10, HR₁₀ 3.9 [2.3,...
G17 ✓ 1b: supported (ledger visibility: R 0.25, HR₁₀ 11.5 ...
+22 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

“Ideas” are hashed marker rules, not semantic units, and $\hat{R}$ depends on the rule. Field vs contagion rests on HR₁₀: a bursty common trigger (a web page or human prompt reaching several agents at once) would mimic it, and copying from tools or the web is invisible. Round 1’s call-start visibility undercounted regime-I exposure; 1b uses the ledger. Unread uses are more recent than read ones, so the convergence share is an upper bound. Novelty is relative to non-holdout text only. The H25 content-dial link (partial $\rho = 0.41$ given room size; daily $\rho = 0.28$) is post hoc and shares its source text with $\hat{R}$. Periods with rooms of 4 cannot show tails.

ROUND 2

Fit a mixture of idea-level R to explain the heavy tail. Predict $\hat{R}$ across the room split and merge before looking. Label numeric claims true or false to compare misinformation with its corrections.

- **Essence.** how far does a new idea or claim spread in an agent swarm, and what sets it: exposure, salience or a shared field?
- **H34-R1** Heterogeneous branching: fit a gamma mixture over idea-level R (one extra parameter) and test whether it removes the excess of trees with five or more agents and beats the beta-binomial on day-ahead log score.
- **H34-R2** Natural experiments for idea spread: the NE15 room split and the 05-04 merge, with the branching ratio and the per-pair branching per room member predicted from room size before looking (axis E).
- **H34-R3** Claims proper: label numeric claims as true or false (or corrected) with Jev and compare branching ratios of false claims and their corrections (the misinformation question).
- **H34-R4** Semantic ideas: embedding-cluster novelty, so paraphrased ideas count, with the same trees and HR₁₀.

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